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2. Sensor noise is problematic when mapping because it can create fake obstacles and make the robot have false readings. It can also hide an object in which the robot can collide with. This can lead to planning errors and unsafe navigation.
3. We chose an increment of $5e-3$ based on testing our code to make sure that a cell is marked as occupied. If it's too small then the mapping may not detect objects, and if it's too big then it may create fake obstacles.
4. We added the threshold of .5 and this is to make sure that cells are marked consistent and are marked as obstacles. If it's too low then it might let the robot think that there's obstacles in the way, and if it's too big then the robot may not see the obstacles and lead to collision.
5. One strategy is to make an online replanning where the robot can update continuously its occupancy grid. If an object that was unexpected on the current path then it'll rerun the path planning algorithm from the current position. You can also add a reactive behavior that'll make the robot move off its path until it's safe to resume.
6. You can use RRT instead of Dijkstra's algorithm, but RRT makes a feasible path that isn't necessarily the shortest path. It might look rand and jagged too, while A* or Dijkstra's is more likely to produce a more smooth and optimal path.
7. We spent well over 18 hours programming this lab.



